

Chemistry 30 — Unit II

Thermochemical Changes

Lesson 3 — Enthalpy

Part A: What Is Enthalpy and Why It Exists

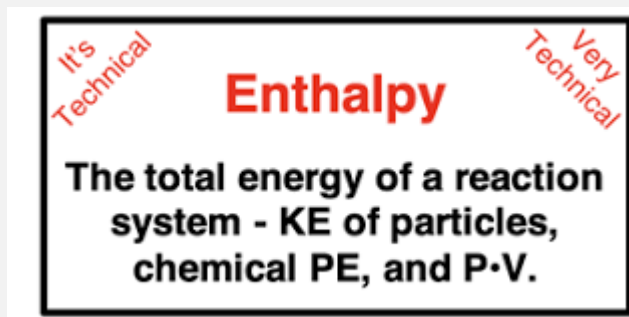
Big Idea

Chemists introduced the concept of enthalpy because chemical reactions always involve energy changes, and those changes needed to be described in a consistent and measurable way. In most real chemical situations—open containers, reactions in air, or reactions in solution—pressure remains constant. Enthalpy is the energy bookkeeping system that works under those conditions.

Students often assume energy is something stored inside a substance like mass or volume. In reality, energy is associated with particle arrangement and the forces between particles. When bonds break and new bonds form, energy must either be absorbed or released. Enthalpy allows chemists to track that transfer reliably.

Understanding enthalpy means understanding energy flow. When a reaction releases energy to its surroundings, the system's enthalpy decreases. When a reaction absorbs energy, the system's enthalpy increases. These changes are not abstract—they explain why fuels burn hot, why cold packs feel cold, and why some reactions must be heated continuously to proceed.

Throughout this lesson, remember that enthalpy is not just a symbol (ΔH). It represents a measurable energy change tied directly to chemical change.



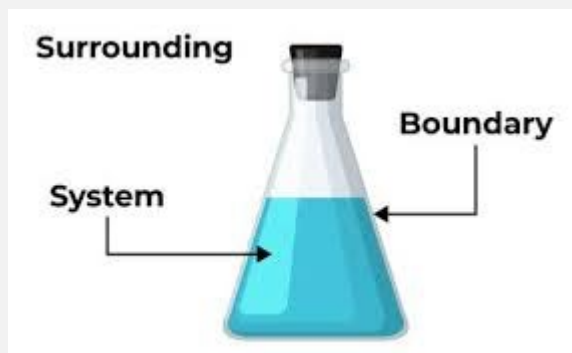
Part B – Understanding Energy Change

🔑 Big Idea 1 – Systems and Surroundings

In thermochemistry, we cannot talk meaningfully about energy unless we first decide what we are paying attention to. Chemists do this by defining a system. The system is the specific part of the universe that we choose to study. It might be a chemical reaction in a beaker, a dissolving salt, or a fuel burning in an engine. Everything outside that chosen boundary is called the surroundings.

This distinction is not arbitrary. Energy does not disappear or appear from nowhere. It is transferred between the system and the surroundings. By clearly defining the system, we can track where energy goes and where it comes from. Without this boundary, terms like ‘released’ or ‘absorbed’ have no meaning.

Students often struggle here because it feels abstract. A helpful way to think about it is to imagine drawing a circle around the reaction you care about. Inside the circle is the system. Everything else — the solution, the container, the air — is the surroundings. Energy always crosses that boundary.



Big Idea 2 - Meaning and Sign of ΔH

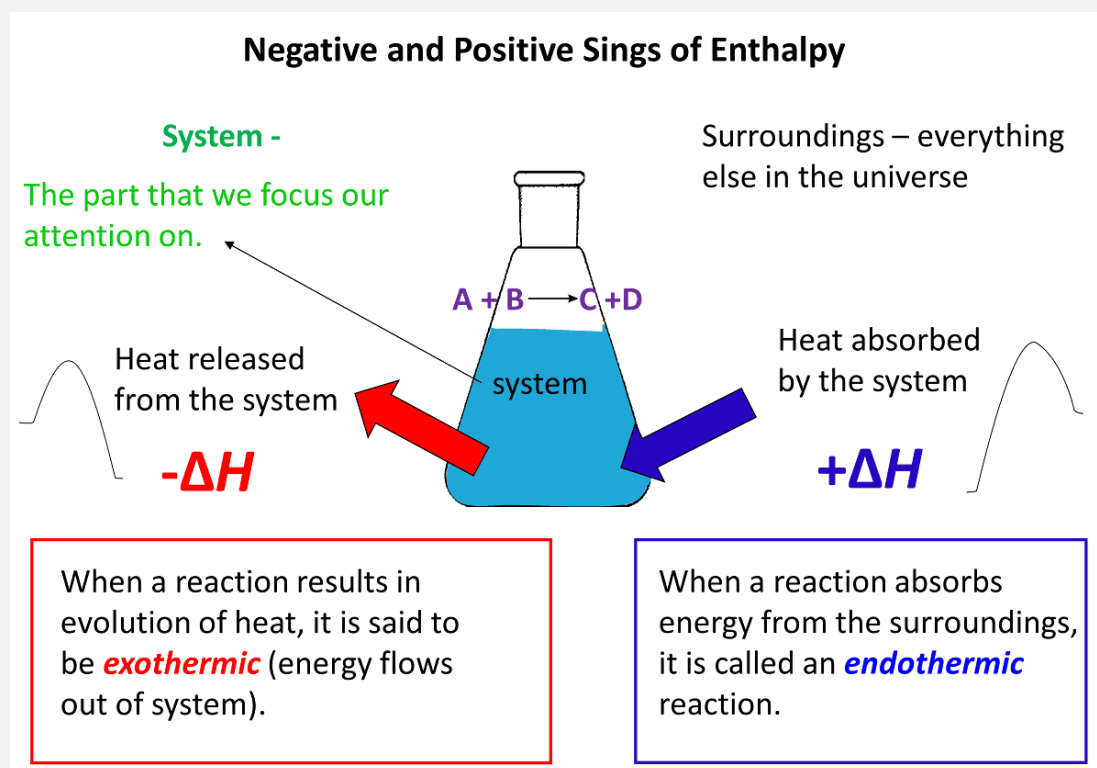
Once a system is defined, we can describe its energy change using enthalpy, symbolized by H . The change in enthalpy, ΔH , tells us how much energy is transferred between the system and the surroundings during a process.

The sign of ΔH is one of the most common sources of confusion for students. The key rule is this: the sign of ΔH always describes the energy change of the system, not the surroundings.

If ΔH is negative, the system has lost energy. That energy was released to the

surroundings, which usually causes the surroundings to warm up. This is called an exothermic process. If ΔH is positive, the system has gained energy. That energy was absorbed from the surroundings, often causing the surroundings to cool. This is called an endothermic process.

A common misconception is to associate negative ΔH with 'cold' and positive ΔH with 'hot.' This is incorrect. A reaction with a negative ΔH can make the surroundings hot because energy is flowing out of the system. Always ask: where is the energy going? That question will guide the sign correctly.



Part C: Standard Enthalpy Changes (ΔH°)

🔑 Big Idea

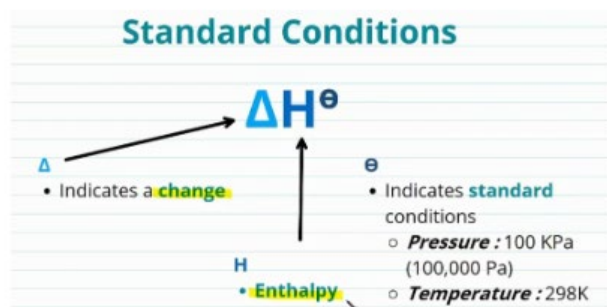
When chemists talk about the energy change of a reaction, they are rarely referring to just any conditions. Instead, they rely on a carefully defined reference framework called standard conditions. Standard enthalpy changes allow scientists to compare energy

changes between reactions in a consistent, meaningful way, regardless of where or how the reaction is studied.

Standard conditions are not chosen arbitrarily. They represent a common baseline: a temperature of 25 °C, a pressure of 100 kPa, and substances in their most stable physical states. By fixing these conditions, chemists remove unnecessary variables and ensure that energy data can be compared, combined, and reused with confidence.

The symbol $\Delta H^\ominus$ is used to indicate that an enthalpy change is measured under standard conditions. The small circle ($^\ominus$) is not decoration—it is a signal that the value only applies when those specific conditions are met. Forgetting this detail is a common source of confusion for students, especially when values are later used in Hess's Law calculations.

- **Pressure:** 100 kPa (1 bar)
- **Temperature:** 298 K (25°C)
- **Concentration (if in solution):** 1 mol dm⁻³
- **Physical state:** All substances must be in their standard physical states



Why Standard Enthalpy Changes Matter

Without standard enthalpy changes, thermochemistry would be fragmented and inconsistent. One lab might measure energy changes at high pressure, another at low temperature, and the results could not be compared or combined. Standard enthalpy values act like a shared language, allowing energy data from different sources to fit together logically.

Types of Standard Enthalpy Changes

Several specific types of standard enthalpy changes are used repeatedly in Chemistry 30. Each describes a different kind of process, but all follow the same rules about conditions and sign conventions.

For example, the standard enthalpy of formation refers to the energy change when one mole of a compound is formed from its elements in their standard states. The standard

enthalpy of combustion refers to the energy released when one mole of a substance burns completely in oxygen under standard conditions.

Students often confuse these definitions by focusing only on the reaction equation and ignoring the wording. It is essential to read each definition carefully. The phrase 'one mole' and the requirement for standard states are not optional details—they define the quantity.

Standard Enthalpy Change of ...	Definition	Symbol	Exothermic/Endothermic
Reaction	The enthalpy change when the reactants in the stoichiometric equation react to give the products under standard conditions	$\Delta H_r^\ominus$	Both
Formation	The enthalpy change when one mole of a compound is formed from its elements under standard conditions	$\Delta H_f^\ominus$	Both
Combustion	The enthalpy change when one mole of a substance is burnt in excess oxygen under standard conditions	$\Delta H_c^\ominus$	Exothermic
Neutralisation	The enthalpy change when one mole of water is formed by reacting an acid and alkali under standard conditions	$\Delta H_{\text{neut}}^\ominus$	Exothermic

Copyright © Save My Exams. All Rights Reserved

Connecting Standard Enthalpy Changes to Thermochemistry

Standard enthalpy changes serve as the building blocks for more complex energy calculations. In later lessons, you will use these values to determine unknown enthalpy changes using Hess's Law. The reliability of those calculations depends entirely on understanding what $\Delta H^\ominus$ represents and how it is defined.

If you can explain, in words, what a given standard enthalpy change means physically, you are far less likely to make sign errors or misuse values in calculations. Thermochemistry rewards conceptual clarity just as much as mathematical accuracy.

Part D: Calculating Enthalpy Changes

Big Idea

In this section, we move from defining enthalpy to using it quantitatively. Chemistry 30 expects students not only to recognize whether a process is endothermic or exothermic, but also to calculate the energy change associated with a chemical or physical process. This requires careful attention to system boundaries, sign conventions, and proportional reasoning.

A common source of confusion is the idea that ΔH is a fixed number regardless of scale. In reality, enthalpy changes scale directly with the amount of substance reacting. This is why thermochemical equations are so powerful: they connect energy change directly to the mole ratios in a balanced equation.

When calculating enthalpy changes, always begin by identifying the reference reaction and its associated ΔH value. Ask yourself: how many moles does this ΔH correspond to? Then compare that to the amount of substance actually involved in the situation you are analyzing.

For example, if a reaction releases -200 kJ when 2.00 mol of reactant are consumed, then consuming 1.00 mol releases -100 kJ , while consuming 4.00 mol releases -400 kJ . The sign of ΔH does not change — only the magnitude does.

Throughout this section, pay careful attention to units and signs. Many errors on diploma-style questions come not from misunderstanding the chemistry, but from losing track of whether energy is being absorbed or released.

$$Q = mc\Delta T$$

HEAT ENERGY (J) Q = m (MASS OF WATER (g)) c (SPECIFIC HEAT CAPACITY) ΔT (TEMPERATURE CHANGE)

$$\Delta H = \frac{Q}{\text{moles}}$$

ENTHALPY CHANGE (J/mol) ΔH = Q (HEAT ENERGY (J)) / moles

Part E: Thermochemical Equations & Scaling Enthalpy

🔑 Big Idea

In this part of the lesson, we focus on how enthalpy changes are written, interpreted, and scaled within chemical equations. Up to this point, you have learned that ΔH represents the energy change of a system during a process. Now, we connect that idea directly to balanced chemical equations. This step is crucial because Chemistry 30 does not treat energy changes in isolation—energy is always tied to a specific chemical change occurring in a specific amount.

A thermochemical equation is a balanced chemical equation that includes an enthalpy change. This equation tells us not only what substances react and form, but also how much energy is absorbed or released when the reaction occurs exactly as written. It is important to understand that the ΔH value is linked to the coefficients in the equation. These coefficients represent amounts of substance, measured in moles, and the enthalpy change corresponds to those exact amounts.

Students often make the mistake of thinking that ΔH is an intrinsic property of a substance, similar to density or melting point. This is not correct. Enthalpy change depends on how much reaction takes place. If more reactant is consumed, more energy is transferred. If less reaction occurs, less energy is involved. This is why scaling reactions requires scaling the enthalpy change by the same factor.

Consider a reaction that releases 100 kJ of energy when 1 mol of a reactant reacts. If 2 mol of that reactant react under the same conditions, the energy released is not still 100 kJ—it is 200 kJ. The reaction has occurred twice as many times at the particle level, so

twice as much energy is transferred. This direct proportionality between amount and energy is at the heart of thermochemical calculations.

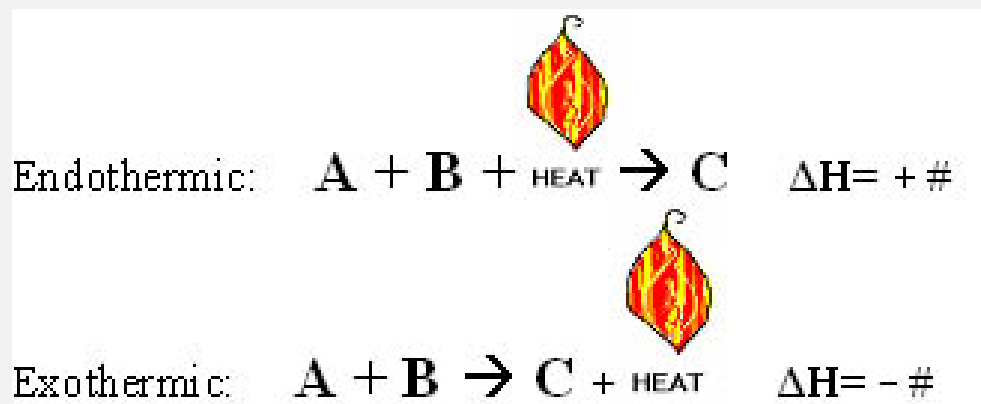
The sign of ΔH must always be interpreted carefully. A negative ΔH indicates that energy is released from the system to the surroundings. This means the reaction is exothermic. A positive ΔH indicates that energy is absorbed by the system from the surroundings, making the reaction endothermic. When scaling a reaction, the sign of ΔH remains the same unless the reaction itself is reversed.

Reversing a thermochemical equation reverses the direction of energy flow. If a reaction releases energy when it proceeds forward, the reverse reaction must absorb that same amount of energy. As a result, when a chemical equation is reversed, the sign of ΔH changes but its magnitude stays the same. This idea will become especially important when you begin using Hess's Law later in the unit.

Another common source of confusion is the placement of energy in a thermochemical equation. Sometimes energy is written as a term on the reactant side or product side instead of being written as ΔH . Both representations describe the same physical reality. Energy written as a product indicates an exothermic reaction, while energy written as a reactant indicates an endothermic reaction.

It is important to remember that thermochemical equations are not just symbolic. They describe real particle-level events. Bonds breaking require energy, while bonds forming release energy. The overall enthalpy change reflects the balance between these two processes. If more energy is released during bond formation than is absorbed during bond breaking, the reaction is exothermic. If more energy is absorbed than released, the reaction is endothermic.

As you move forward in Thermochemistry, you will repeatedly use thermochemical equations to calculate energy changes, compare reaction pathways, and evaluate efficiency. Developing a strong understanding of how enthalpy scales with chemical reactions now will make the remainder of this unit far more manageable and conceptually clear.



Part F: Comparing Enthalpy Changes and Reaction Pathways

Big Idea

In this final part of Lesson 3, we focus on how enthalpy values are compared and interpreted to make meaningful conclusions about chemical reactions. Up to this point, you have learned how to define enthalpy, determine whether a process is endothermic or exothermic, and scale enthalpy changes based on the amount of reaction. Now, we step back and ask a broader question: how do chemists use enthalpy data to compare reactions and evaluate which processes are more energy-efficient or more energetically favourable?

When comparing enthalpy changes, it is essential to remember that ΔH values are always relative. They describe the energy change associated with a specific reaction under specific conditions. A reaction with a larger negative ΔH releases more energy than one with a smaller negative ΔH , but that does not automatically mean it is 'better' or 'preferred' in all situations. Context matters, especially when energy efficiency, safety, and control are considered.

Students often assume that reactions which release the most energy are always the most desirable. In reality, extremely exothermic reactions can be difficult to control and may pose safety risks. For example, the combustion of fuels releases large amounts of energy, which is useful for heating and power generation, but must be carefully managed to prevent damage or explosions. This is why understanding enthalpy is not just an academic exercise—it has direct practical and societal importance.

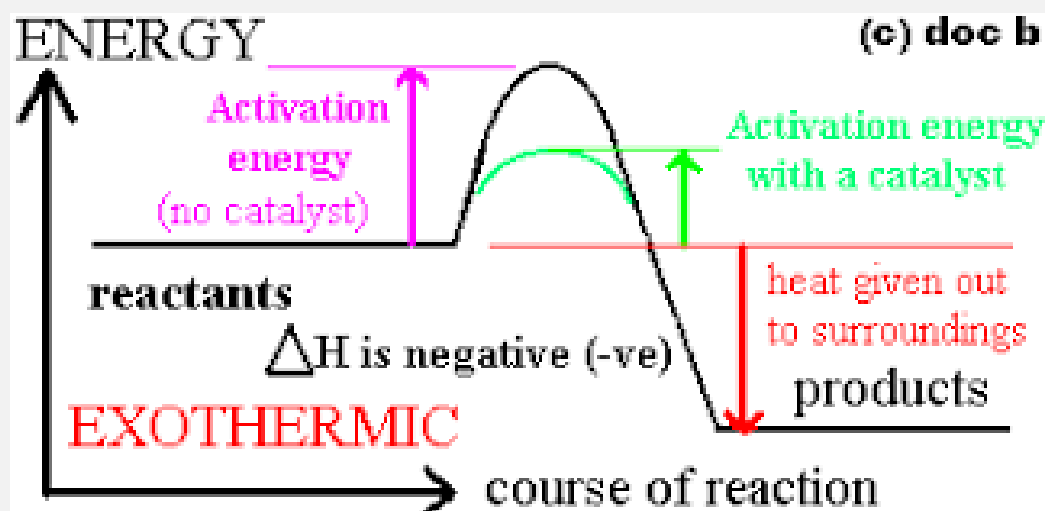
Another important idea in this section is the concept of reaction pathways. Chemical reactions do not simply jump from reactants to products. Instead, they proceed through a series of molecular events involving bond breaking and bond forming. Different pathways between the same reactants and products can have different energy profiles, even though the overall enthalpy change remains the same.

This idea sets the stage for Hess's Law, which you will study next. Hess's Law is based on the principle that enthalpy is a state function. This means that the total enthalpy change depends only on the initial and final states of a system, not on the path taken between them. Whether a reaction occurs in one step or many steps, the overall ΔH remains unchanged.

At this point, it is important to pause and reflect on a common misconception. Students sometimes think that changing the pathway of a reaction changes the total energy released or absorbed. While catalysts and intermediates can change the rate of a reaction or the activation energy required, they do not change the overall enthalpy change. Energy accounting always balances between the start and end points.

By learning how to compare enthalpy changes and reason about reaction pathways, you are developing the skills needed to analyze real chemical systems. These skills are essential for understanding industrial chemistry, environmental processes, and energy technologies. More importantly for Chemistry 30, they prepare you to confidently tackle Hess's Law problems and multi-step enthalpy calculations that appear frequently on diploma-style assessments.

As you move forward, keep returning to the central idea of this lesson: enthalpy tells a story about energy flow. Your job as a chemist is to interpret that story carefully, using both numerical values and conceptual reasoning to draw accurate conclusions.



Student Practice Questions

1. Define enthalpy in your own words and explain why it is useful in chemistry.
2. Identify the system and surroundings for a chemical reaction occurring in a coffee-cup calorimeter.
3. A reaction has a ΔH of -250 kJ . Explain what this sign and value indicate.
4. Calculate the energy released when a reaction with $\Delta H = -125 \text{ kJ}$ occurs twice.
5. Write a thermochemical equation for the combustion of methane.
6. Explain why reversing a reaction changes the sign of ΔH .
7. Compare two reaction pathways that have different activation energies but the same ΔH .
8. Explain how enthalpy changes relate to energy efficiency in industry.

Worked Solutions

1. Enthalpy is the total energy content of a system that can be transferred as heat at constant pressure.
2. The system is the reacting chemicals; the surroundings include the water and calorimeter.
3. The negative sign means energy is released; 250 kJ of heat flows to the surroundings.
4. Doubling the reaction doubles the energy: -250 kJ.
5. $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g}) \quad \Delta H = -890 \text{ kJ}$
6. Reversing a reaction reverses energy flow, changing the sign of ΔH .
7. Different activation energies affect rate, not total enthalpy change.
8. Enthalpy data helps industries design reactions that maximize useful energy while minimizing waste.

Diploma-Style Practice Set

This mixed-response practice set mirrors the style and cognitive demand of Alberta Chemistry 30 diploma questions. It includes multiple-choice and written-response items. Complete all questions before consulting the worked solutions.

Part A: Multiple Choice (Select the Best Answer)

1. Which statement best describes enthalpy (H)?
 - A. Energy stored only in chemical bonds
 - B. Total energy of a system that can be transferred as heat at constant pressure
 - C. Rate of energy transfer during a reaction
 - D. Energy required to start a reaction
2. A reaction has $\Delta H = +85 \text{ kJ}$. This means:
 - A. 85 kJ of energy is released to the surroundings
 - B. The surroundings lose 85 kJ of energy
 - C. 85 kJ of energy is absorbed by the system
 - D. The reaction cannot occur
3. Which change will reverse the sign of ΔH for a reaction?
 - A. Increasing temperature
 - B. Adding a catalyst
 - C. Reversing the reaction
 - D. Doubling the reactants
4. Two reactions have identical ΔH values but different activation energies. This means:
 - A. They release different amounts of energy
 - B. One reaction is faster than the other
 - C. One reaction is endothermic
 - D. Energy conservation is violated
5. In a coffee-cup calorimeter, which is the system?
 - A. The thermometer
 - B. The surrounding air
 - C. The reacting chemicals
 - D. The cup and water

Part B: Written Response

6. Explain the difference between an exothermic and an endothermic reaction using both energy flow and ΔH sign.

7. A reaction has $\Delta H = -120 \text{ kJ}$. Calculate the total energy change if the reaction occurs three times. Explain your reasoning.

8. Describe how enthalpy changes are represented in thermochemical equations and why coefficients matter.

9. Explain why a reaction pathway can change without changing the overall ΔH of a reaction.

10. Discuss one real-world example where understanding enthalpy changes is important for safety or efficiency.

Worked Solutions

Multiple Choice Answers:

1. B
2. C
3. C
4. B
5. C

Written Response Sample Solutions:

6. Exothermic reactions release energy to the surroundings and have a negative ΔH , while endothermic reactions absorb energy and have a positive ΔH .
7. If the reaction occurs three times, the energy change is -360 kJ because enthalpy scales with reaction amount.
8. Thermochemical equations include ΔH values that correspond to the reaction as written; changing coefficients changes the energy proportionally.
9. Different pathways may require different activation energies, but the total enthalpy change depends only on initial and final states.
10. In industrial fuel combustion, controlling enthalpy changes improves efficiency and prevents overheating or explosions.

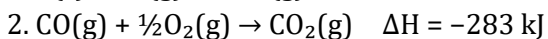
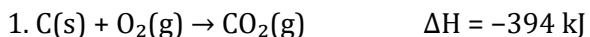
Lesson 3 Challenge Question

(Bridge to Hess's Law)

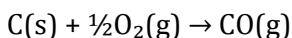
This challenge question is designed to prepare you for the next major idea in Thermochemistry: Hess's Law. Do not worry if this feels difficult at first. The goal is to begin thinking about enthalpy changes as quantities that can be combined and compared logically.

Challenge Question

A student is given the following thermochemical equations:



The student wants to determine the enthalpy change for the reaction:



- Explain why this enthalpy change cannot be measured directly.
- Using reasoning (not just calculation), determine the ΔH value for this reaction.
- Explain how this process illustrates the principle behind Hess's Law.

Worked Solution

a) This reaction cannot be measured directly because carbon monoxide is unstable and reacts readily with oxygen to form carbon dioxide. As a result, the reaction occurs in steps rather than as a single, isolated process that can be measured in a calorimeter.

b) The target reaction produces CO(g) , while both given equations produce $\text{CO}_2\text{(g)}$. To isolate the formation of CO , we must conceptually subtract the second reaction from the first.

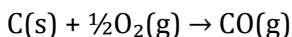
Reaction 1 releases 394 kJ when carbon forms CO_2 .

Reaction 2 releases 283 kJ when CO forms CO_2 .

If forming CO_2 from C releases 394 kJ, and converting CO to CO_2 releases 283 kJ, then forming CO from C must release the remaining energy:

$$\Delta H = -394 \text{ kJ} - (-283 \text{ kJ}) = -111 \text{ kJ}$$

Therefore, the enthalpy change for:



is $\Delta H = -111 \text{ kJ}$.

c) This illustrates Hess's Law because it shows that the total enthalpy change depends only on the initial and final states of the system, not on the pathway taken. By combining known thermochemical equations, we can determine the enthalpy change for a reaction that cannot be measured directly.

This type of reasoning—reversing reactions, adding or subtracting equations, and tracking energy—will be formalized and practiced extensively in the next lesson on Hess's Law.

Lesson 3 Exit Ticket

Enthalpy → Hess's Law

This exit ticket is designed to check your understanding of the core ideas from Lesson 3 and to prepare your thinking for the next lesson on Hess's Law. Answer each question carefully. Clear reasoning matters more than perfect wording.

Part A: Core Understanding

1. In one or two sentences, explain what enthalpy (ΔH) represents in a chemical reaction. Your answer should clearly refer to both the system and the surroundings.
2. A reaction has a ΔH value of -120 kJ.
 - a) Is the reaction endothermic or exothermic?
 - b) Describe the direction of energy flow.
 - c) Explain what the negative sign tells you physically.

Part B: Scaling and Reasoning

3. If the reaction in Question 2 occurs twice, determine the total energy change. Explain why your answer makes sense in terms of enthalpy scaling.

Part C: Bridge to Hess's Law

4. Some reactions cannot be measured directly in a calorimeter.
 - a) Give one reason why this might be the case.
 - b) Explain how using known enthalpy changes from other reactions can still allow us to determine the overall ΔH .
5. The challenge problem in this lesson required combining information from two thermochemical equations to find a third.

In your own words, explain why the total enthalpy change depends only on the starting materials and final products, not on the reaction pathway.

Self-Reflection

6. After completing this lesson, how confident do you feel about explaining enthalpy changes?
☐ Very confident ☐ Somewhat confident ☐ Still unsure